■ CHVN Paper Reviewer

Multi-Agent Scientific Paper Review

qwen3:8b · May 03, 2026 21:21

File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	qwen3:8b
Target Publisher	IEEE

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Editorial Decision

MINOR REVISION

Weighted Score: 3.8 / 5.0
Confidence: 83%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**5.0**	Excellent	■■■■■
Manuscript Structure	**2.5**	Weak	■■■■■
Methodology & Equations	**4.0**	Good	■■■■■
Statistical Analysis	**4.5**	Excellent	■■■■■
Results & Evidence	**4.2**	Good	■■■■■
Discussion & Conclusions	**3.5**	Good	■■■■■
Figures & Tables	**3.5**	Good	■■■■■
Scientific Writing	**3.5**	Good	■■■■■
References & Citations	**3.0**	Acceptable	■■■■■
Ethics & Transparency	**1.5**	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **5.0 / 5.0** | Confidence: 95%

Strengths:

- [TITLE] Specific and informative, clearly reflects the contribution of the physics-informed EANN framework.
- [ABSTRACT] Structured format with clear background, objective, methods, results, and conclusions.
- [KEYWORDS] Comprehensive and domain-specific, with a mix of general and specific terms.
- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.

StructureReviewer

Manuscript structure and logical coherence

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical focus on physics-informed EANN framework
- Experimental validation with seismic base excitation
- Reproducibility emphasized through framework description

Weaknesses:

- Missing literature review section
- Redundant content in section_vi and section_vii
- Research question/hypothesis not explicitly declared

Major Comments:

- The manuscript lacks a literature review section, which is critical for establishing novelty and context in IEEE publications.
- Sections VI and VII are redundant and should be merged or removed.
- Research question/hypothesis is not explicitly declared, which weakens the logical flow.

Minor Comments:

- Section titles are inconsistent and not clearly labeled (e.g., 'section_vi' instead of 'Section VI').
- Some figures are referenced but not fully described in the text.

Recommendations:

- Add a comprehensive literature review section to contextualize the work.
- Merge or remove redundant sections VI and VII.
- Explicitly declare the research question or hypothesis in the introduction.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- Clear mathematical formulation of the structural model and optimization problem
- Robust statistical validation with nonparametric tests and confidence intervals
- Use of real-world experimental data with detailed preprocessing steps

Weaknesses:

- Equations are not consistently introduced or referenced in the text

- Lack of explicit justification for the choice of experimental benchmark
- Limited discussion on the computational complexity and scalability of the EANN framework

Major Comments:

- The authors should clarify the rationale for selecting the five-story building prototype as the experimental benchmark.

Minor Comments:

- Equations should be consistently introduced and referenced in the text for better reproducibility.
- The discussion on the computational efficiency and scalability of the EANN framework is limited.

Recommendations:

- Provide a more detailed explanation of the physical constraints and their implementation in the EANN framework.
- Clarify the rationale for the selection of the five-story building prototype as the experimental benchmark.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **4.5 / 5.0** | Confidence: 90%

Strengths:

- Use of Kruskal-Wallis and post-hoc Wilcoxon with Holm correction for non-normal data
- Bootstrap confidence intervals for robustness
- Coefficient of variation and RMSE/MSE for practical significance
- Pareto front and hypervolume indicators for multi-objective analysis

Weaknesses:

- No specific p-values reported for statistical tests
- No mention of power analysis or sample size justification beyond 30 runs

Major Comments:

- The manuscript should report p-values for Kruskal-Wallis and post-hoc tests to fully support statistical inference.

Minor Comments:

- Consider explicitly stating the number of resamples used in bootstrap confidence intervals for transparency.

Recommendations:

- Report p-values for all statistical tests to ensure clarity and reproducibility.
- Provide justification for the choice of 30 runs, including any power analysis or simulation-based reasoning.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Equations are numbered sequentially and referenced in the text
- Figures and tables are referenced in the text before they appear
- The methodology is clearly described with mathematical formulations

Weaknesses:

- Some figures lack axis labels and units
- Some tables lack units and statistical significance indicators
- Some equations are referenced but not fully defined in the text

Major Comments:

- Equations (11), (13), and (16) are referenced but not fully defined in the text, which may lead to confusion for readers.
- Figures 3, 4, and 5 lack axis labels and units, which reduces their clarity and self-explanatory nature.

- Table X lacks units and statistical significance indicators, which reduces its interpretability.

Minor Comments:

- Figure 6 does not indicate error bars or uncertainty for comparison.
- The text does not specify which metrics are included in Table X.

Recommendations:

- Define equations (11), (13), and (16) in full in the text to ensure clarity.
- Add axis labels and units to Figures 3, 4, and 5 to improve clarity.
- Include units and statistical significance indicators in Table X to improve interpretability.

ResultsReviewer

Results validity and empirical support

Score: **4.2 / 5.0** | Confidence: 80%

Strengths:

- Clear presentation of MSE, RMSE, CV, and hypervolume metrics
- Baseline comparisons with GA and PSO are explicitly detailed
- Statistical validation methods are described with appropriate rigor

Weaknesses:

- Negative results are not explicitly reported or acknowledged
- Some quantitative results (e.g., CV, hypervolume) are not tied to specific figures or tables
- The exact methodology for computing the Pareto front and its comparison to baselines is not sufficiently detailed

Major Comments:

- The manuscript lacks explicit reporting of negative results, which is critical for transparency and reproducibility.

Minor Comments:

- Some quantitative results (e.g., CV, hypervolume) are not clearly linked to figures or tables.

Recommendations:

- Include explicit reporting of negative results or limitations in the results section.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear experimental validation
- Reproducibility emphasized
- Technical novelty in framework

Weaknesses:

- Discussion lacks comparison with prior work
- Overgeneralization in implications
- Future work is vague

Major Comments:

- The Discussion section lacks a direct comparison with prior IEEE publications, which is critical for establishing novelty.
- The conclusions overgeneralize the implications of the results without sufficient justification.
- Future work is not clearly specified and lacks concrete directions.

Minor Comments:

- The Discussion could benefit from addressing unexpected findings in more detail.

• The conclusion section could be more concise and avoid repetition of the abstract.

Recommendations:

• Include a direct comparison with prior IEEE publications in the Discussion section.

• Clarify the theoretical and practical implications without overgeneralizing.

• Specify concrete future work directions that build on the current findings.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

• Clear technical description of the EANN framework

• Detailed experimental setup and validation

• Statistical validation methods are well described

Weaknesses:

• Overuse of passive voice

• Unclear acronyms

• Vague quantifiers without numerical support

Major Comments:

• The manuscript lacks clear definitions for key acronyms such as EANN, PSO, and GA on first use.

• Passive voice is overused and should be minimized for clarity and directness.

• Vague quantifiers like 'significant' and 'many' should be replaced with numerical values for precision.

Minor Comments:

• Some sentences are run-on and could be split for clarity.

• The paragraph structure could be improved for better flow.

Recommendations:

• Define all acronyms on first use.

• Revise passive voice constructions to active voice where possible.

• Replace vague quantifiers with specific numerical values.

• Split long sentences for improved readability.

ReferencesReviewer

References quality, recency, and relevance

Score: **3.0 / 5.0** | Confidence: 80%

Strengths:

• Recent references (2021-2023) are included

• Citations are consistent with IEEE style

Weaknesses:

• Missing seminal foundational works in structural dynamics

• Lack of key references on seismic design regulations

Major Comments:

• The references section lacks seminal foundational works in structural dynamics and seismic design regulations, which are critical for the field.

• The manuscript should include more recent and foundational references to support the claims made in the introduction and methodology.

Minor Comments:

• The reference [34] is incomplete and should be corrected for clarity and completeness.

Recommendations:

- Include seminal foundational works in structural dynamics and seismic design regulations.
- Ensure all references are complete and properly formatted according to IEEE standards.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.5 / 5.0** | Confidence: 90%

Strengths:

- The methodology and experimental validation are well described, with clear technical contributions and mathematical rigor.

Weaknesses:

- Ethics and limitations reporting are insufficient, which is critical for IEEE standards.

Major Comments:

- The manuscript lacks essential ethical reporting, including ethical approval, informed consent, data availability, and conflict of interest declarations. These are required for IEEE publication standards.

Minor Comments:

- Reproducibility is not addressed, as data and code availability are not stated.

Recommendations:

- Authors must provide ethical approval details, data availability statements, and conflict of interest declarations.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] The manuscript lacks a literature review section, which is critical for establishing novelty and context in IEEE publications.

[2] Sections VI and VII are redundant and should be merged or removed.

[3] Research question/hypothesis is not explicitly declared, which weakens the logical flow.

[4] The authors should clarify the rationale for selecting the five-story building prototype as the experimental benchmark.

[5] The manuscript should report p-values for Kruskal-Wallis and post-hoc tests to fully support statistical inference.

[6] Equations (11), (13), and (16) are referenced but not fully defined in the text, which may lead to confusion for readers.

[7] Figures 3, 4, and 5 lack axis labels and units, which reduces their clarity and self-explanatory nature.

[8] Table X lacks units and statistical significance indicators, which reduces its interpretability.

[9] The manuscript lacks explicit reporting of negative results, which is critical for transparency and reproducibility.

[10] The Discussion section lacks a direct comparison with prior IEEE publications, which is critical for establishing novelty.

[11] The conclusions overgeneralize the implications of the results without sufficient justification.

[12] Future work is not clearly specified and lacks concrete directions.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] Section titles are inconsistent and not clearly labeled (e.g., 'section_vi' instead of 'Section VI').

[2] Some figures are referenced but not fully described in the text.

[3] Equations should be consistently introduced and referenced in the text for better reproducibility.

[4] The discussion on the computational efficiency and scalability of the EANN framework is limited.

[5] Consider explicitly stating the number of resamples used in bootstrap confidence intervals for transparency.

[6] Figure 6 does not indicate error bars or uncertainty for comparison.

[7] The text does not specify which metrics are included in Table X.

[8] Some quantitative results (e.g., CV, hypervolume) are not clearly linked to figures or tables.

[9] The Discussion could benefit from addressing unexpected findings in more detail.

[10] The conclusion section could be more concise and avoid repetition of the abstract.

[11] Some sentences are run-on and could be split for clarity.

[12] The paragraph structure could be improved for better flow.

Specific Improvement Recommendations

1. Add a comprehensive literature review section to contextualize the work.
2. Merge or remove redundant sections VI and VII.
3. Explicitly declare the research question or hypothesis in the introduction.
4. Provide a more detailed explanation of the physical constraints and their implementation in the EANN framework.
5. Clarify the rationale for the selection of the five-story building prototype as the experimental benchmark.
6. Report p-values for all statistical tests to ensure clarity and reproducibility.
7. Provide justification for the choice of 30 runs, including any power analysis or simulation-based reasoning.

8. Define equations (11), (13), and (16) in full in the text to ensure clarity.
9. Add axis labels and units to Figures 3, 4, and 5 to improve clarity.
10. Include units and statistical significance indicators in Table X to improve interpretability.
11. Include explicit reporting of negative results or limitations in the results section.
12. Include a direct comparison with prior IEEE publications in the Discussion section.